



ELK406E

Distribution of Electrical Energy

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Introduction to Distribution Systems

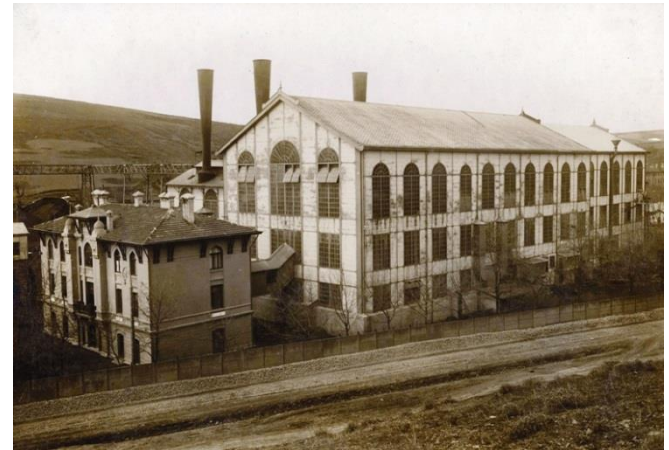


Brief History of Electric Power

- Incandescent Lamp, Thomas Edison, 1879
- First power station Pearl Street, Manhattan, *Thomas Edison*, 1882 marked the beginning of the electric utility industry. At Pearl Street, ***dc generators*** were driven by steam engines to supply an initial load of 30 kW for 110 V incandescent lighting.
- *Frank J. Sprague* produced ***dc motor*** for Edison systems, 1884
- *William Stanley* developed commercially practical ***transformer***, 1885
- *Nikola Tesla* introduced the ***polyphase ac system***, as well as the ***polyphase induction*** and ***synchronous motor***, which made evident the advantages of polyphase versus single-phase systems, 1888

in **TURKEY**

- The first electric generator was a 2 kW dynamo connected to the water mill installed in Tarsus in 1902
- The first bigger power plant was installed in Istanbul, Silahtarğa in 1913 (Thermal plant, 17 MW power generation capacity)
- The installed capacity had been 33 MW till the date of the founding of the Republic of Turkey after the war of independence



in **TURKEY**



- 99.9% of Turkey's population benefit from the electrical energy
- In 1948, Çatalağzı Thermal Power Plant was put into service and the network of Istanbul had been reinforced by a 154 kV transmission line installed in 1952
- Today, the country had entirely been netted with interconnected transmission lines meanwhile thousands of transformers of various capacity and types were equipped.

Utility Deregulation

The objectives of electric utility restructuring are

- to increase competition
- to decrease regulation
- to obtain lower consumer prices

The electric power industry is becoming market-driven.

The consumer will have a choice as to from which generator they purchase power



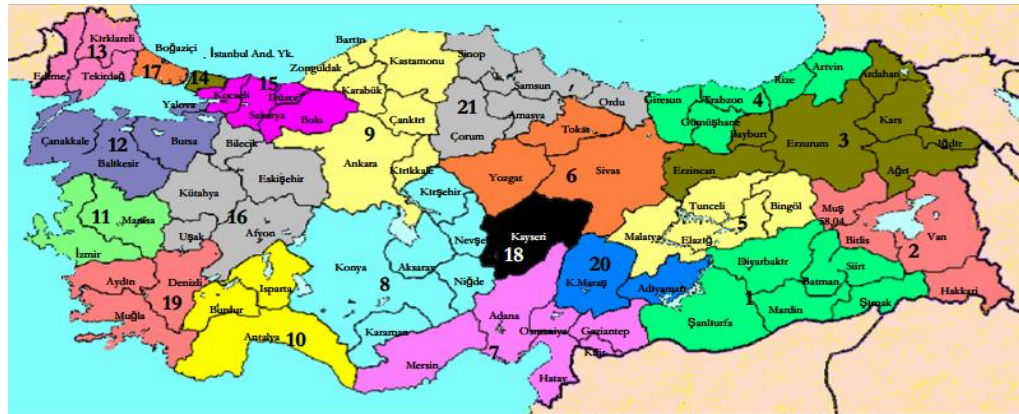
Restructuring in Turkey - 1

- TEK (public-owned monopoly for G, T & D.)
- 1993, TEK split to TEIAS and TEDAS
- 2001, Energy Market Regulatory Authority
- 2002, EMRA (EPDK) Licensing and Tariff Regulations
- 2004, Privatization Strategy Document
- 2009, generation & distribution privatization starts



Restructuring in Turkey - 2

- 2013, process finalized in 21 geographical regions



- 2013, distribution & retail sales are also separated,
- 2015, customers are free to choose suppliers,
- eligibility threshold expected to be decreased to zero,
- Organized Industrial Zones (OIZ) can obtain distribution license

1.1 Tarihi Gelişim



TEAS: Turkish Electricity Generation and Transmission Co. Inc.,
TEDAS: Turkish Electricity Distribution Co. Inc.,

Transmission: The transport of electricity through lines higher than 36 kV,

Distribution: The transport of electricity through 36 kV or lower lines;



The Energy Market Regulatory Authority ("EMRA")



REPUBLIC OF TURKEY
ENERGY MARKET
REGULATORY AUTHORITY

ELECTRICITY MARKET LAW

Law No.: 5346 : 4628
Adoption date : 20/2/2001
Published in the Official Gazette : Bis No. 24335 dated 3/3/2001

Purpose, Scope, Definitions

ARTICLE 1– (1) — The purpose of this Law is to ensure the development of a financially sound and transparent electricity market operating in a competitive environment under provisions of civil law and the delivery of sufficient, good quality, low cost and environment- friendly electricity to consumers and to ensure the autonomous regulation and supervision of this market.

(2) The scope of this law covers generation, transmission, distribution, wholesale, retailing and retailing services, import, export of electricity; rights and obligations of all real persons and legal entities directly involved in these activities; establishment of Electricity Market Regulatory Authority| and determination of operating principles of this authority; and the methods to be employed for privatization of electricity generation and distribution assets.

11. Generation: The transformation of energy resources into electricity in generating facilities,

12. Transmission: The transport of electricity through lines higher than 36 kV,

13. Distribution: The transport of electricity through 36 kV or lower lines;

14. Wholesale: The sale of electricity for resale;

15. Retail: The sale of electricity to consumers;

16. Retail Sale Service: Excluding the sale of electricity and/or capacity, the services provided by companies holding retail sale licenses to consumers;

17. Consumer: All eligible or non-eligible consumers, purchasing electricity for their own needs;

18. Supplier: (Amended by the Law no. 5784 dated 9 July 2008 /Article 1) Generation companies, autoproducers, autoproducer groups, wholesale companies and retail sale companies providing electricity and/or capacity;²

19. Eligible consumer: Any real person or legal entity that has the liberty to choose its supplier, due to its consumption of more electricity than the amount set by the Board and/or its direct connection to the transmission system;

20. Non-eligible consumer: Any real person or legal entity that can purchase electricity energy and/or capacity only from retail sale companies or from a distribution company holding a retail sale license in its region;

21. Generation Company: Any legal entity, except for autoproducers and autoproducer groups, engaged in generation of electricity and the sale of the electricity it has generated;

22. Distribution Company: Any legal entity engaged in electricity distribution in a certain geographical region;

23. Wholesale Company: Any legal entity engaged in the wholesale, import, export, trade of electricity energy and/or capacity and the sale of the same to the eligible consumers;

24. Retail Sale Company: Any legal entity engaged in import of electricity and/or capacity and retail sale to consumers, excluding those directly connected to the transmission system, and in providing retail sale services to consumers;

25. Autoproducer: Any legal entity engaged in electricity generation primarily for its own needs;

26. Autoproducer Group: Any legal entity engaged in electricity generation primarily for needs of its affiliates;

27. Facility: Plant and equipment installed to perform the functions of generation, transmission or distribution of electricity;

28. Distribution System: Electricity distribution facilities and grid operated and/or owned by a distribution company in its designated region

TURKEY'S ELECTRICITY DISTRIBUTION COMPANIES

■ Private



Distribution Company: Any legal entity engaged in electricity distribution in a certain geographical region;

Turkish Electricity Distribution System (in 2022)

	33 kV	15 kV	10.5 kV	6.3 kV	0.4kV	Total
Overhead Line (km)	400193.7	15352.3	195.1	465.8	592579.5	1008786.4
Underground Cable (km)	75340.7	3288.8	4148.5	925.8	177216.2	260919.9

MV distribution Voltage levels: 6kV, 10 kV, 15 kV, 33, kV

Transformer s	MV/MV	MV/LV	Total
Total number	1629	547335	548964
Power (MVA)	14290.2	210229.7	224519.9

Number of customers in 2022:
48,557,252

Poles	MV	LV
Metal pole	2,571,389	5,856,607
Concrete pole	1,539,891	3,140,040
Wood-pole	579,156	4,363,452
Total	4,690,436	13,360,099

	2022 (GWh)
Transmission Losses	5,417
Distribution Losses	23,650
Net energy consumption	286,576

2018

TÜRKİYE KURULU GÜCÜNÜN KURULUŞLARA DAĞILIMI Share of the Producers in Turkey's Installed Capacity	
KURULUŞLAR Producers	KURULU GÜÇ Installed Capacity (MW)
Serbest Üretim Şirketleri Generation Companies	53.856
EÜAŞ	18.496
Yap-İşlet (Yİ) Build-Own-Operate (BOO)	6.102
Lisanssız Üretim Şirketleri Unlicensed Generation Companies	5.375
İşletme Hakkı Devri (İHD) Transfer of Operation Rights (TOR)	3.387
Yap-İşlet-Devret (YİD) Build-Operate-Transfer (BOT)	1.346
TOPLAM / Total	88.562

2022

Türkiye Kurulu Gücünün Kuruluşlara Göre Dağılımı

KURULUŞ	MW
EÜAŞ	21.183
Serbest Üretim Şti.	70.784
Lisanssız Santraller	8.697
İşletme Hakkı Devri	3.275
Yap İşlet Devret	116
TOPLAM	104.055

Türkiye Elektrik Üretiminin Kaynaklara Göre Dağılımı - 2022

Türkiye Kurulu Gücünün Kaynaklara Göre Dağılımı

KAYNAKLAR	MW
Hidrolik	31.586
Doğalgaz	25.519
Rüzgar	11.396
Linyit + Taş Kömürü	11.032
İthal Kömür	10.374
Güneş	9.486
Biyokütle	1.922
Jeotermal	1.691
Sıvı Yakıtlar *	662
Atık Isı	387
TOPLAM	104.055

(*) : Sıvı Yakıtlar (Fuel Oil+Motorin+ Nafta+Asfaltit)

KAYNAKLAR	MWh
Doğalgaz	71.264.752
Hidrolik	66.752.993
İthal Kömür	63.167.939
Linyit + Taş Kömürü	49.474.275
Rüzgar	34.922.306
Güneş	15.119.137
Jeotermal	11.079.025
Biyokütle	7.937.918
Sıvı Yakıtlar *	5.213.325
Atık Isı	955.903
TOPLAM	325.887.572

(*) : Sıvı Yakıtlar (Fuel Oil+Motorin+ Nafta+Asfaltit)

Kurulu Güç ve Elektrik Üretimi Gelişimi

Developments of Installed Capacity and Electricity Generation

TÜRKİYE				
Yıllar	KURULU GÜÇ		ÜRETİM	
	MW	Değişim %	GWh	Değişim %
1970	2.235		8.623	
1971	2.578	15,35	9.781	13,43
1972	2.711	5,17	11.242	14,93
1973	3.193	17,75	12.425	10,53
1974	3.732	16,90	13.477	8,47
1975	4.187	12,18	15.623	15,92
1976	4.364	4,24	18.283	17,03
1977	4.727	8,32	20.565	12,48
1978	4.869	2,99	21.726	5,65
1979	5.119	5,13	22.522	3,66
1980	5.119	0,00	23.275	3,35
1981	5.538	8,18	24.673	6,00
1982	6.639	19,88	26.552	7,61
1983	6.935	4,47	27.347	3,00
1984	8.462	22,01	30.614	11,95
1985	9.122	7,80	34.219	11,78
1986	10.115	10,89	39.695	16,00
1987	12.495	23,53	44.353	11,73
1988	14.521	16,21	48.049	8,33
1989	15.808	8,87	52.043	8,31
1990	16.318	3,22	57.543	10,57
1991	17.209	5,46	60.246	4,70
1992	18.716	8,76	67.342	11,78
1993	20.338	8,66	73.808	9,60
1994	20.860	2,57	78.322	6,12
1995	20.954	0,45	86.247	10,12
1996	21.249	1,41	94.862	9,99
1997	21.892	3,02	103.296	8,89
1998	23.354	6,68	111.022	7,48
1999	26.119	11,84	116.440	4,88
2000	27.264	4,38	124.922	7,28
2001	28.332	3,92	122.725	-1,76
2002	31.846	12,40	129.400	5,44
2003	35.587	11,75	140.580	8,64
2004	36.824	3,48	150.698	7,20
2005	38.820	5,42	161.956	7,47
2006	40.502	4,33	176.300	8,86
2007	40.836	0,82	191.558	8,65
2008	41.817	2,40	198.418	3,58
2009	44.761	7,04	194.813	-1,82
2010	49.524	10,64	211.208	8,42
2011	52.911	6,84	229.395	8,61
2012	57.058	7,84	239.497	4,40
2013	64.008	12,18	240.154	0,27
2014	69.520	8,61	251.963	4,92
2015	73.150	5,22	261.783	3,90
2016	78.521	7,34	274.407	4,82
2017	85.360	8,71	297.278	8,33
2018	88.562	3,75	303.413	2,06
2019	91.431	3,24	302.552	-0,28
2020	95.964	4,96	304.612	0,68
2021	99.713	3,91	330.806	8,60
2022	104.055	4,35	325.888	-1,49

TEK - TEAŞ - EÜAŞ				
Yıllar	KURULU GÜÇ		ÜRETİM	
	MW	Değişim %	GWh	Değişim %
T E K	1970	1.439	6.273	
	1971	1.764	7.802	24,38
	1972	1.878	9.124	16,95
	1973	2.350	10.258	12,43
	1974	2.834	11.189	9,07
	1975	3.229	12.845	14,80
	1976	3.385	15.454	20,31
	1977	3.685	17.230	11,49
	1978	3.801	17.968	4,28
	1979	3.685	18.934	5,38
	1980	4.051	19.415	2,54
	1981	4.051	20.588	6,04
	1982	4.442	23.243	12,90
	1983	5.543	23.689	1,92
	1984	5.936	26.686	12,65
T E A Ş	1985	7.795	30.249	13,35
	1986	8.789	35.470	17,26
	1987	11.014	39.679	11,87
	1988	12.984	43.014	8,40
	1989	14.240	47.454	10,32
	1990	14.729	52.854	11,38
	1991	15.317	55.461	4,93
	1992	16.800	61.533	10,95
	1993	18.280	67.100	9,05
	1994	18.649	71.943	7,22
E Ü A Ş	1995	18.858	78.195	8,69
	1996	18.905	85.415	9,23
	1997	19.070	90.919	6,44
	1998	19.563	96.075	5,67
	1999	21.119	92.313	-3,92
	2000	21.252	93.234	1,00
	2001	21.063	86.362	-7,37
	2002	21.058	77.332	-10,46
	2003	21.785	63.097	-18,41
	2004	21.790	68.017	7,80
	2005	22.584	73.462	8,00
	2006	23.714	84.530	15,07
	2007	23.875	92.327	9,22
	2008	23.981	97.717	5,84
	2009	24.203	89.454	-8,46
	2010	24.203	95.532	6,80
	2011	24.150	92.351	-3,33
	2012	24.775	90.575	-1,92
	2013	23.782	80.118	-11,55
	2014	21.879	70.469	-12,04
	2015	20.323	55.319	-21,50
	2016	20.145	46.509	-15,93
	2017	20.032	47.094	1,26
	2018	18.496	45.799	-2,75
	2019	19.751	59.350	29,59
	2020	21.588	55.585	-6,34
	2021	21.172	53.524	-3,71
	2022	21.183	46.971	-12,24

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MARKETS

GENERATION

CONSUMPTION

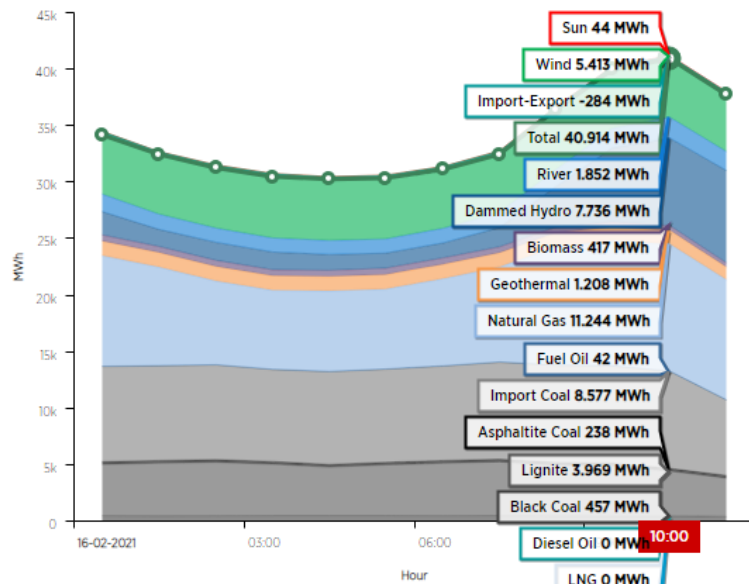
Renewables Support Mechanism

TRANSMISSION

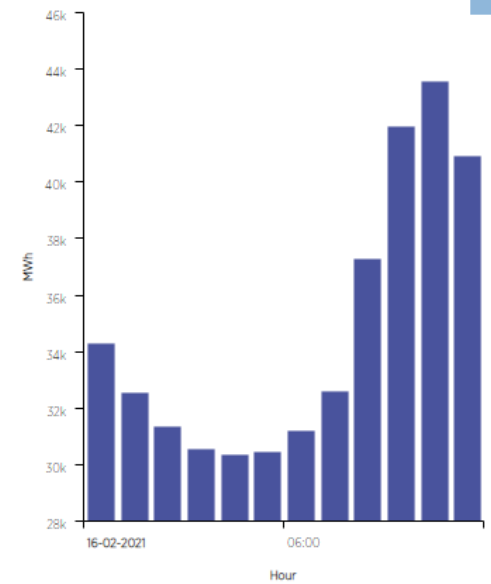
NATURAL GAS Transmission

Natural Gas Continuous Trading
Platform (CTP)

Real-Time Generation ⓘ



Real-Time Consumption ⓘ



Day-Ahead Market ⓘ



Intraday Market ⓘ

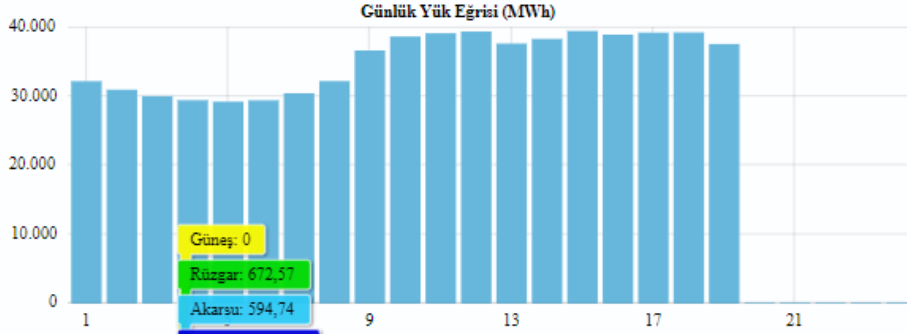


Balancing Power Market ⓘ

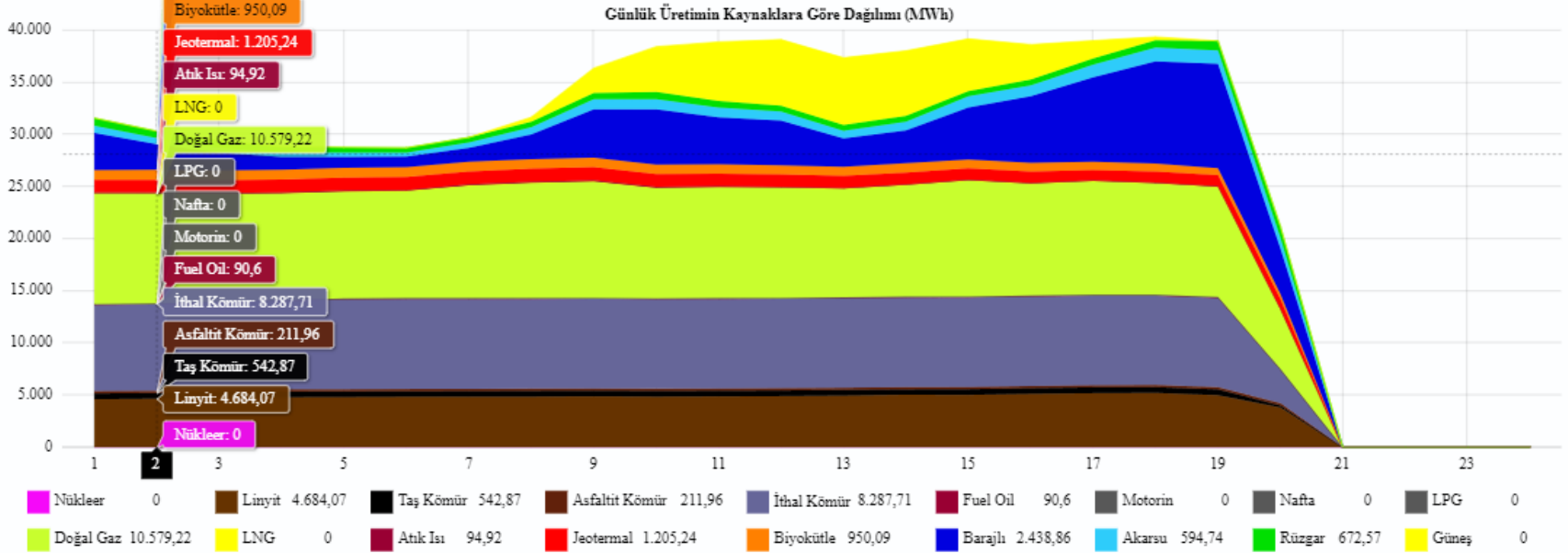
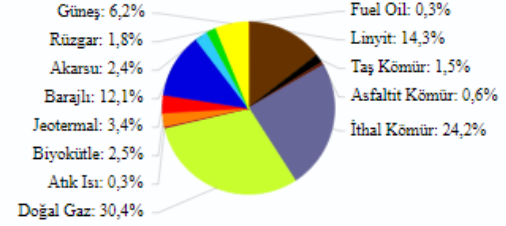


Tarih : * 31-10-2023

Göster



Saatlik Üretim Kaynaklara Göre Dağılımı (MWh)





TEİAŞ

TURKISH
ELECTRICITY
TRANSMISSION
CORPORATION

NET TRANSFER CAPACITIES

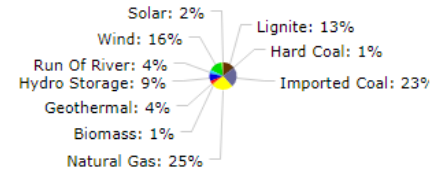
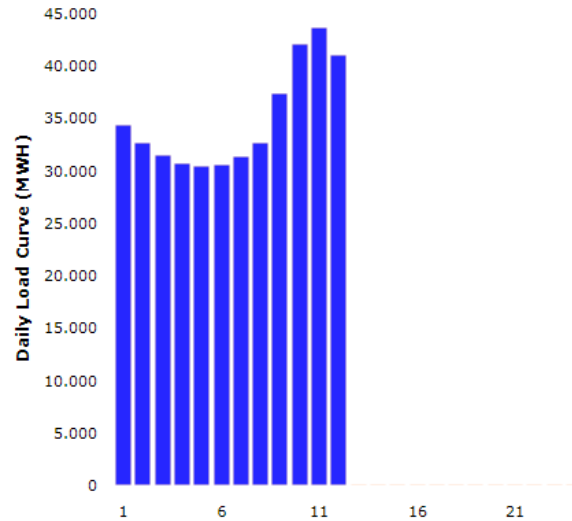
Net Transfer Capacities

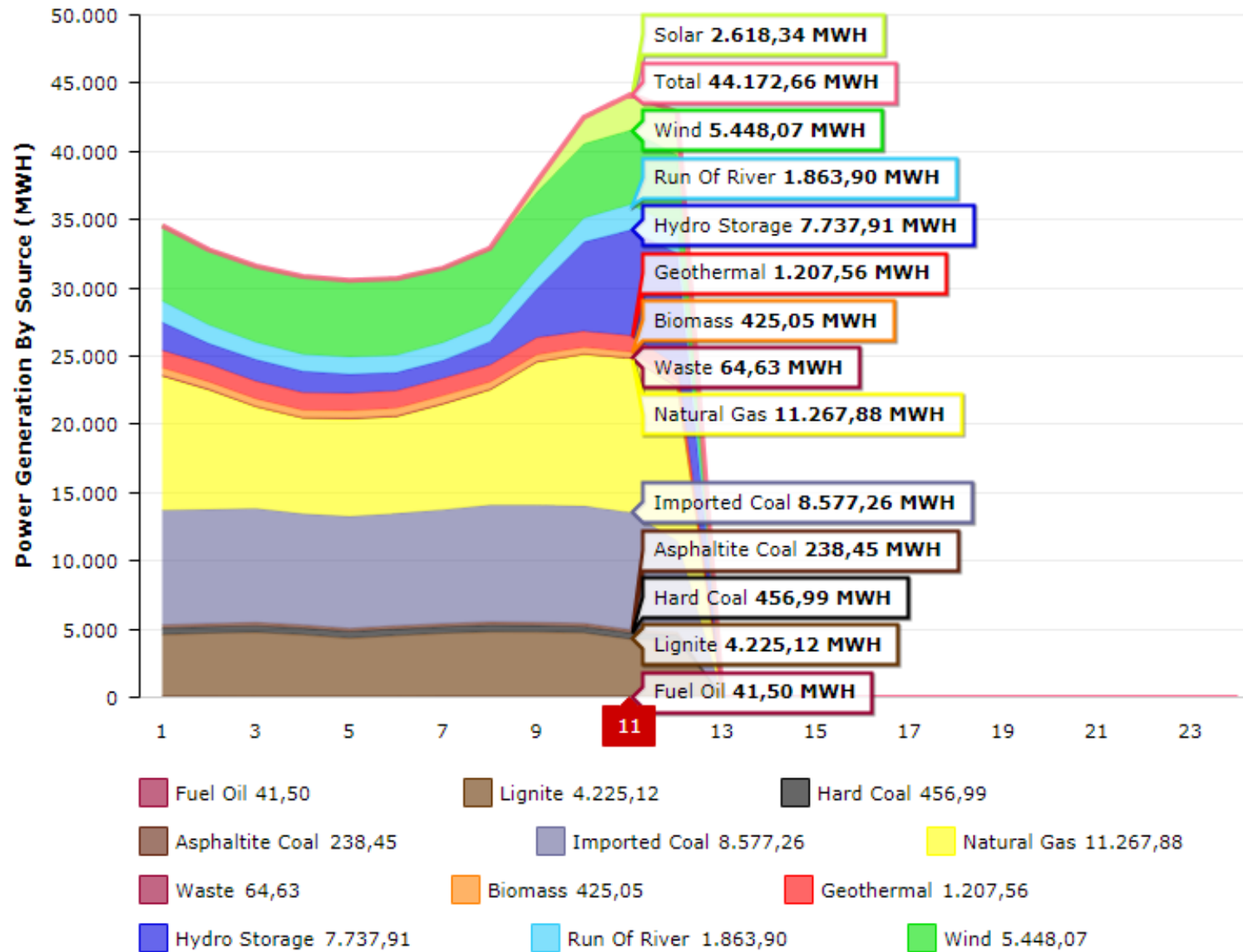


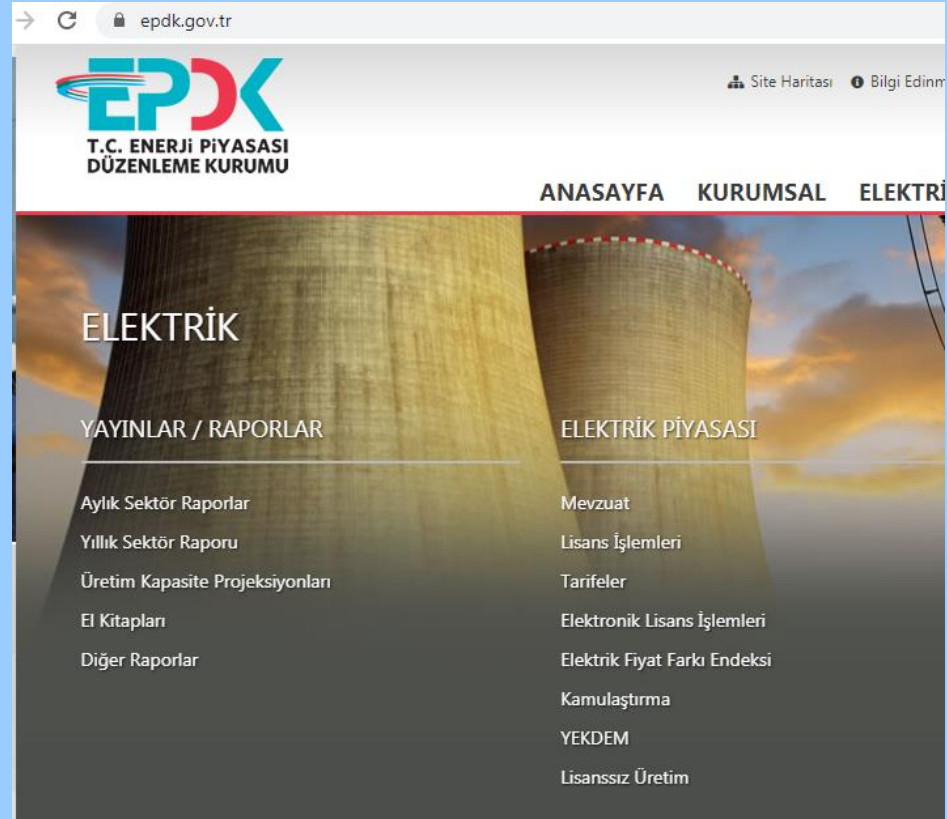
Daily Statistics

Date : *

2021-02-16

[View](#)





in **TURKEY**

- Three-phase transmission system voltages in Turkey are 154 kV and 400 kV.
- There are 400/154, 400/MV and 154/MV step-down transformers in the system.
- The transmission lines operating at 66 kV have been upgraded to transmit power at 154 kV or 400 kV.
- There are also 220 kV transmission lines to interconnect power systems in Turkey, Armenia and Georgia.
- Power transmission losses are 2.5-3 %.

The existing interconnections

- Turkey-Azerbaycan (Nahcivan)

Aralık(Turkey)-Sederek(Nahcivan); 34.5 kV 10 MW

Iğdır(Turkey)-Babek(Nahcivan); 154 kV, double-circuit, 100 MW

- Turkey- Armenia

Kars (Turkey)–Leninakan (Armenia);220 kV, 300 MW

No energy transfer

- Turkey-Georgia

Hopa(Turkey)- Batum(Georgia); 220 kV, 300 MW

- Turkey-Bulgaria

Babaeski(Turkey)-Maritsa East(Bulgaria); 400 kV, 500 MW

- Turkey-Syria

- Turkey-Iraq

PS3(Turkey)-Zakho(IRaq); 400 kV 500 MW (operated at 154 kV)

- Turkey-Iran

Doğubeyazıt(Turkey)-Bazargan(Iran); 154 kV, 100 MW

Modern Power System

A power system can be subdivided into four major parts:

- Generation
- Transmission and Subtransmission
- Distribution
- Loads

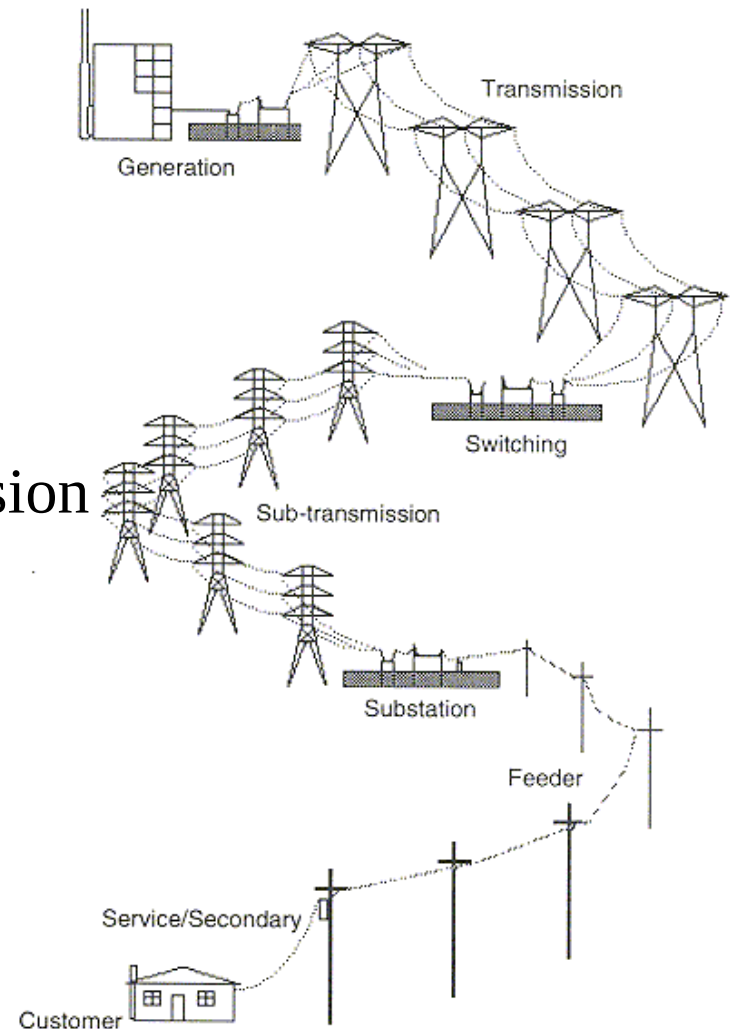
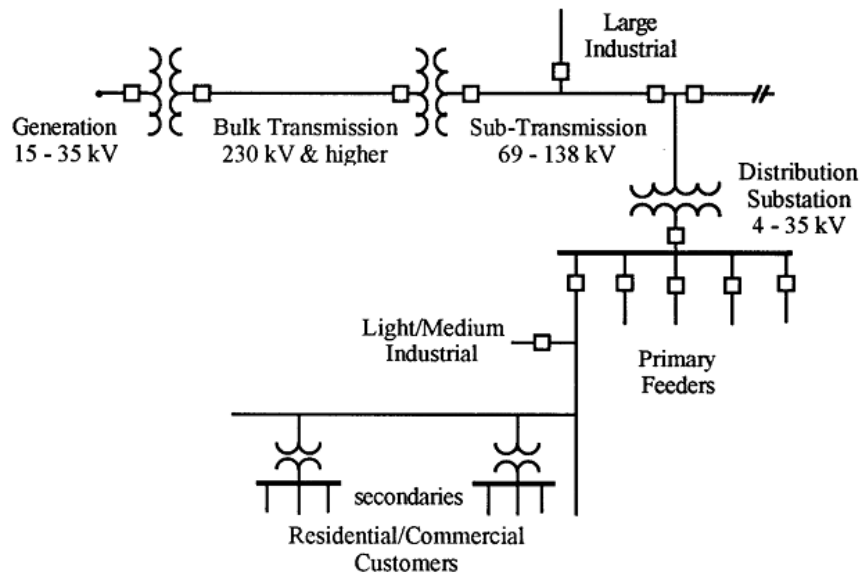
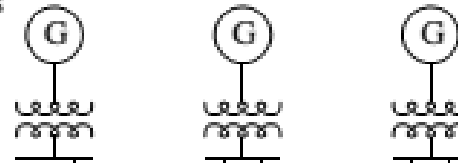
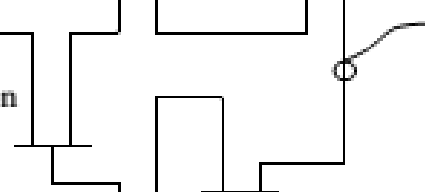


FIGURE 6.1 Representative portion of a typical power system configuration.

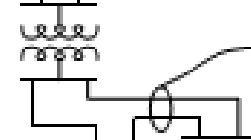
Large Generation
Stations



Bulk Transmission
230-750 kV



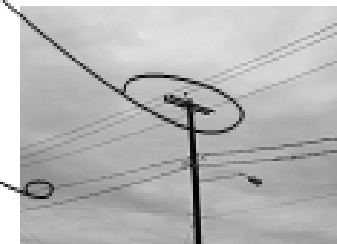
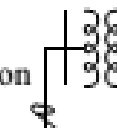
Subtransmission
69-169 kV



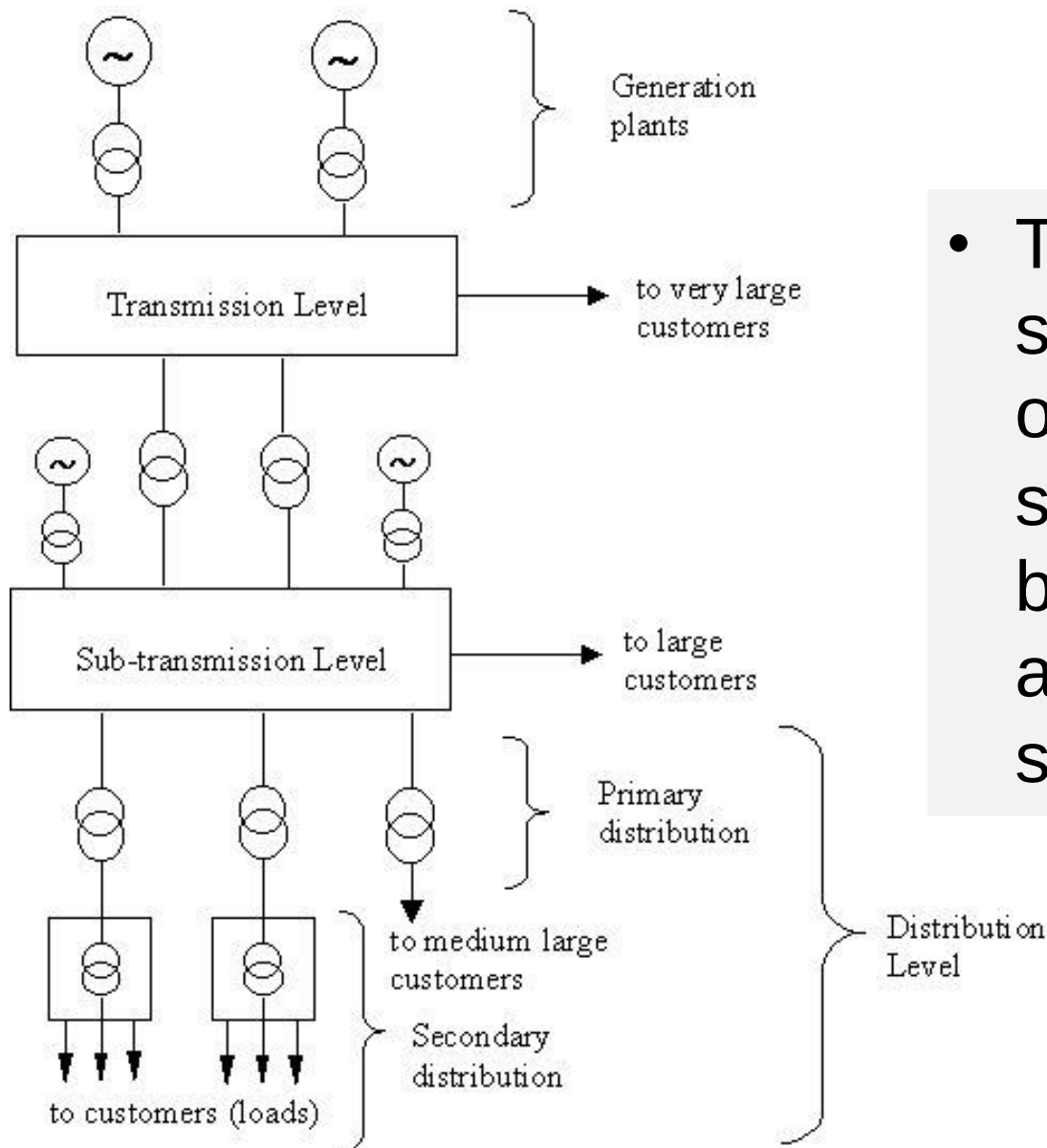
Primary Distribution
4-35 kV



Secondary Distribution
120/240 V



Introduction



- The distribution system is that part of the electric utility system between the bulk power source and the customers' service switches.

Introduction

The distribution system includes the following components

- Subtransmission system
- Distribution substations
- Distribution or primary feeders
- Distribution transformers
- Secondary circuits
- Service drops

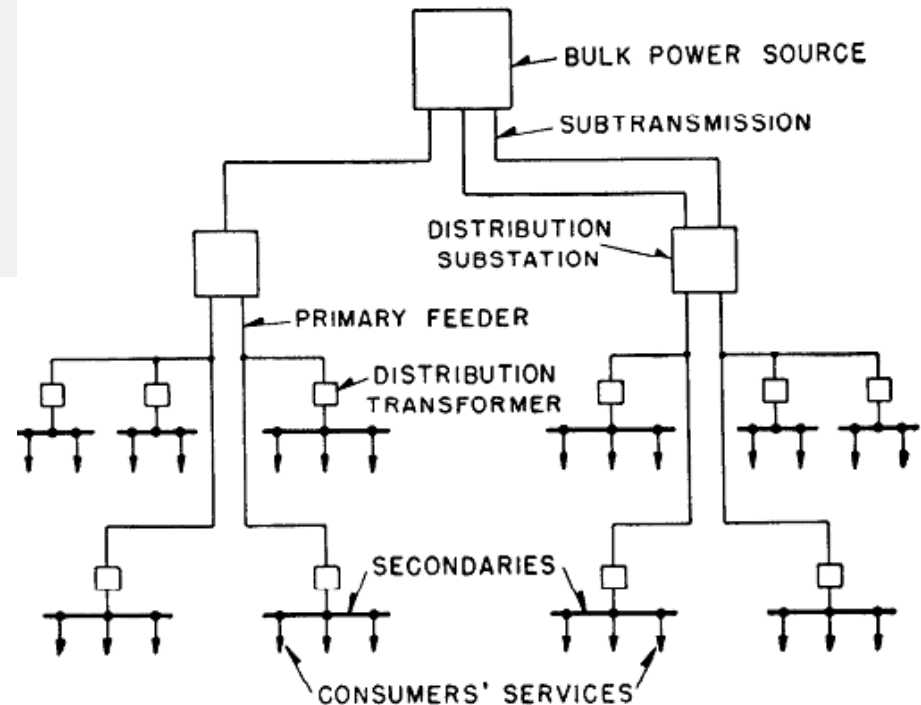


Fig. 1—Typical distribution system showing component parts.

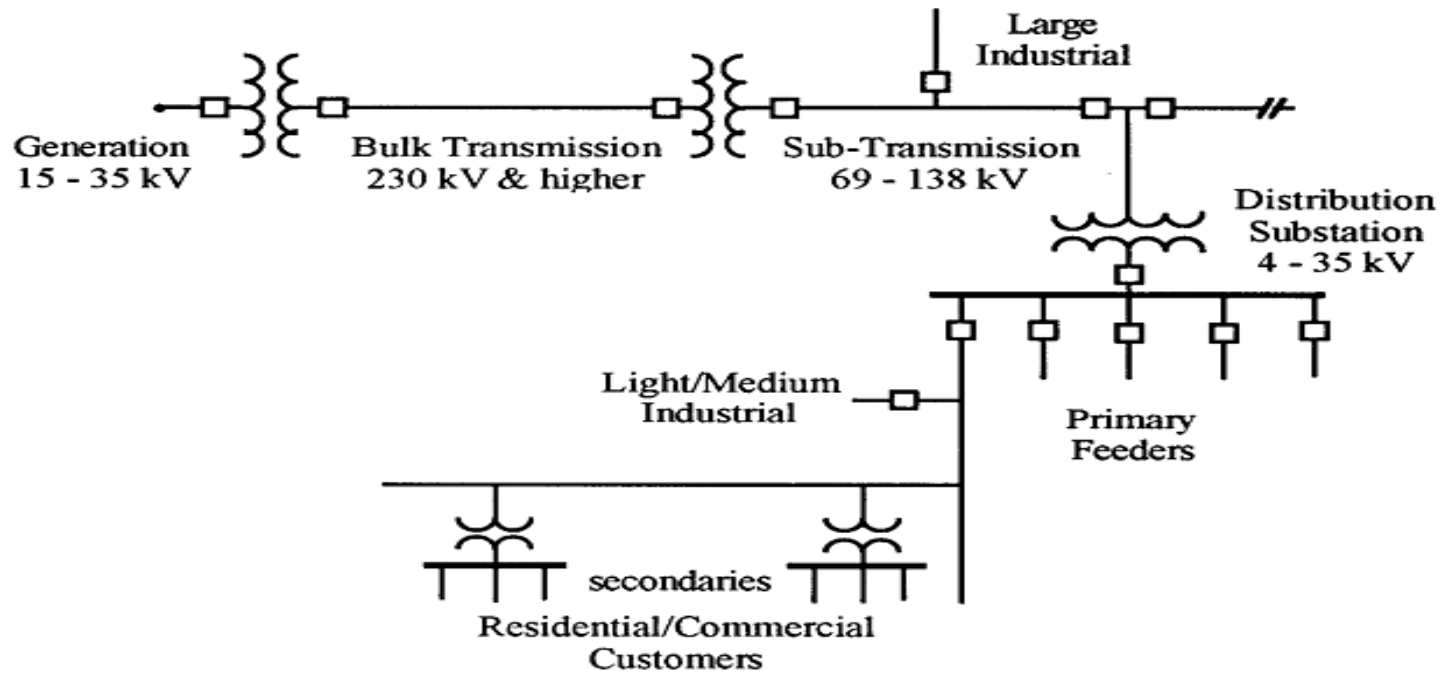
The operating voltages

Voltage class	Nominal voltage
Low	120/240 V (1 phase in US) 220 V (1 phase <i>in Turkey</i>) 380 V (3 phase <i>in Turkey</i>)
Medium	2.4 kV, , 6.9 kV, 20 kV 34.5 kV (<i>in Turkey</i>) , 69 kV
High	115 kV, 138 kV 154 kV (<i>in Turkey</i>) 161 kV, 230 kV
Extra High	345 kV 380 kV (<i>in Turkey</i>) 500 kV, 765 kV

- Distribution system connects the distribution substations to the consumers.
- The primary distribution lines are usually in the range of 4 to 34.5 kV
- The secondary distribution network reduces the voltage for utilization by commercial and residential consumers
- Distribution systems can be overhead or underground



Introduction



- The three-phase primary feeders are usually operated in the range of 4.16 to 34.5 kV
- The three-phase primary feeders distribute energy from the low-voltage bus of the substation to its load center where it branches into three-phase subfeeders and single laterals.

Introduction

- Distribution transformers, in ratings from 10 to 1000 kVA, are usually connected to each primary feeder, subfeeders, and laterals.
- They reduce the distribution voltage to the utilization voltage.
- The secondaries facilitate the path to distribute energy from the distribution transformer to consumers through service drops.

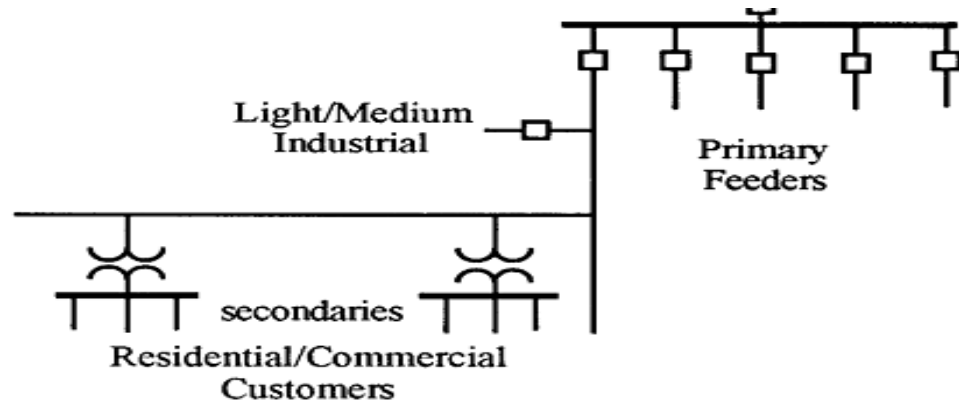


FIGURE 6.1 Representative portion of a typical power system configuration.

Introduction

- The voltage delivered to customers must be within a suitable range, so utilities must regulate the voltage.
- On distribution circuits, voltage drops due to current flowing through the line impedances.
- Primary and secondary voltage drop can be allocated as necessary along the circuit to provide end users with suitable voltage.
- Voltage regulators — in the substation or on feeders — can adjust primary voltage.

- The standards specify the acceptable operational ranges at two locations on electric power systems.
- **Service voltage** is the point where the electrical systems of the supplier and the user are interconnected. (normally at the meter)
Maintaining acceptable voltage at the service entrance is the utility's responsibility
- **Utilization voltage** is the voltage at the line terminals of utilization equipment. This voltage is the facility's responsibility.
- Equipment manufacturers should design equipment which operates satisfactorily within the given limits

The effects of improper voltage regulation

- Improper voltage regulation can cause many problems for end users. Sustained overvoltages or undervoltages can cause the following end-use impacts:
 - Improper or less-efficient equipment operation. For example, lights may give incorrect illumination or a machine may run fast or slow.
 - Tripping of sensitive loads. For example, an uninterruptible power supply (UPS) may revert to battery storage during high or low voltage. This may drain the UPS batteries and cause an outage to critical equipment.

Undervoltages can cause overheating of induction motors.

- For lower voltage, an induction motor draws higher current.
- Operating at 90% of nominal, the fullload current is 10 to 50% higher, and the temperature rises by 10 to 15%.
- With less voltage, the motor has reduced motor starting torque.

Overvoltages can cause equipment damage or failure.

- Equipment can suffer insulation damage.
- Incandescent light bulbs wear out much faster at higher voltages.
- Higher no-load losses in transformers.
- Magnetizing currents are higher at higher voltages.